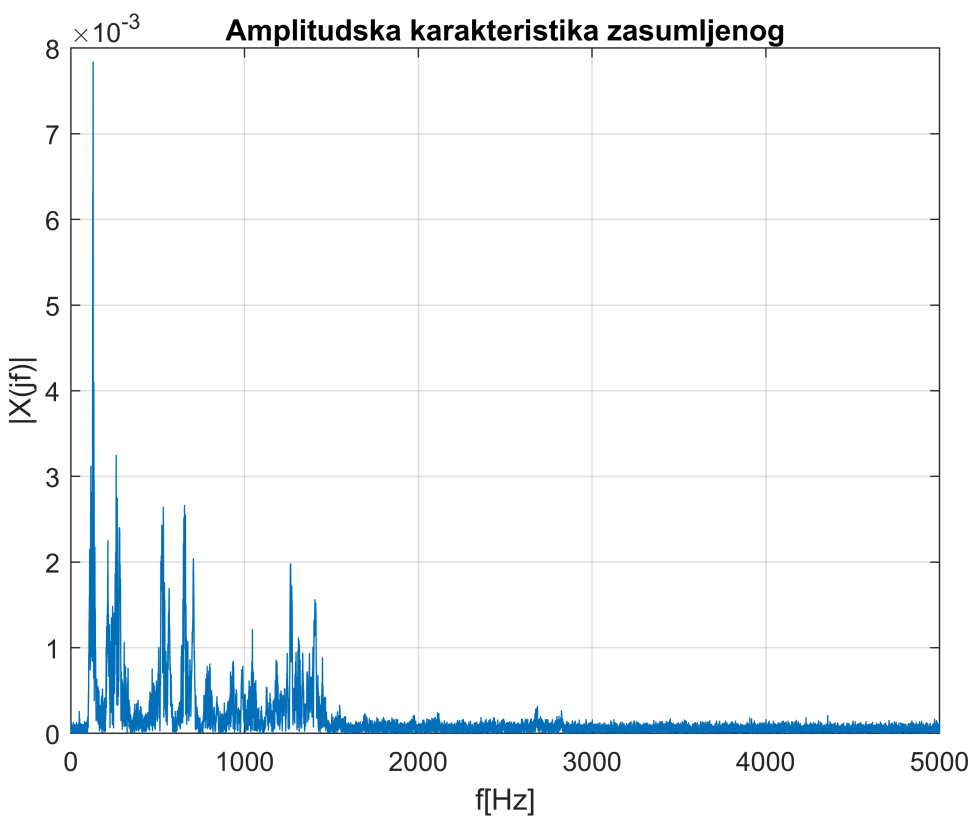
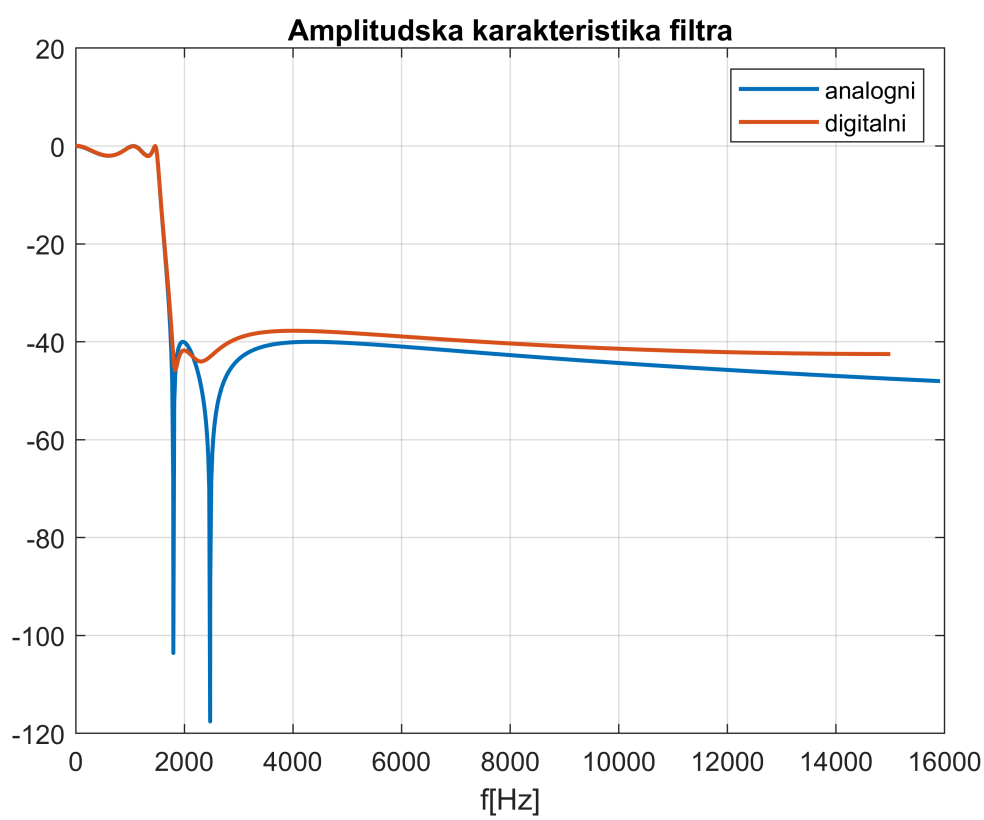
```
clear all; close all; clc;

[s, Fs] = audioread('C:\Users\38166\Desktop\govor.wav');
t = 0 : 1/Fs : (length(s)-1)/Fs;

y1 = s + 0.01*randn(length(s), 1); %zasumljivanje
%audiowrite('C:\Users\38166\Desktop\govor_zasumljen.wav', y1, Fs);
%-------------------------------------------------------------
N = 2^nextpow2(length(y1)); %fft zasumljenog
f1 = 0 : Fs/N : Fs/2;
X = fft(y1, N) / length(y1);
X1 = abs(X(1 : N/2+1));
X1(2 : N/2+1) = 2*X1(2 : N/2+1);

Ns = 2^nextpow2(length(s)); %fft mog glasa
fs = 0 : Fs/Ns : Fs/2;
Xs = fft(s, Ns) / length(s);
Xs1 = abs(Xs(1 : Ns/2+1));
Xs1(2 : Ns/2+1) = 2*Xs1(2 : Ns/2+1);

figure(1);
plot(f1, X1);
xlim([0 5000]);
xlabel('f[Hz]'); ylabel('|X(jf)|');
title('Amplitudska karakteristika zasumljenog'); grid on;
```

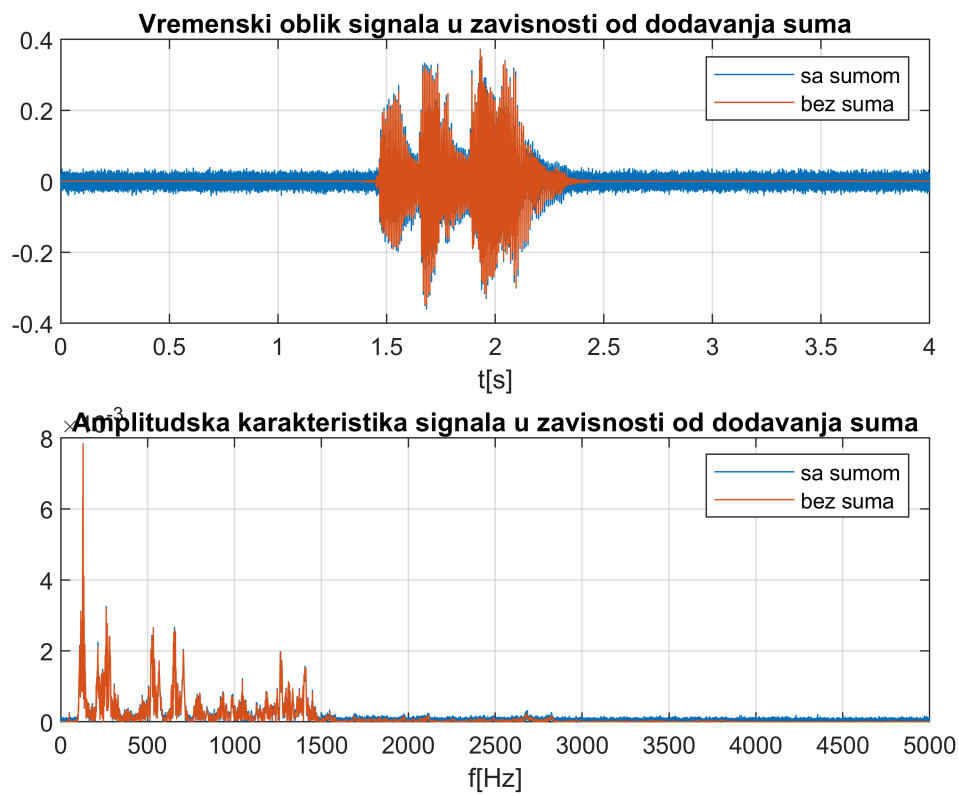


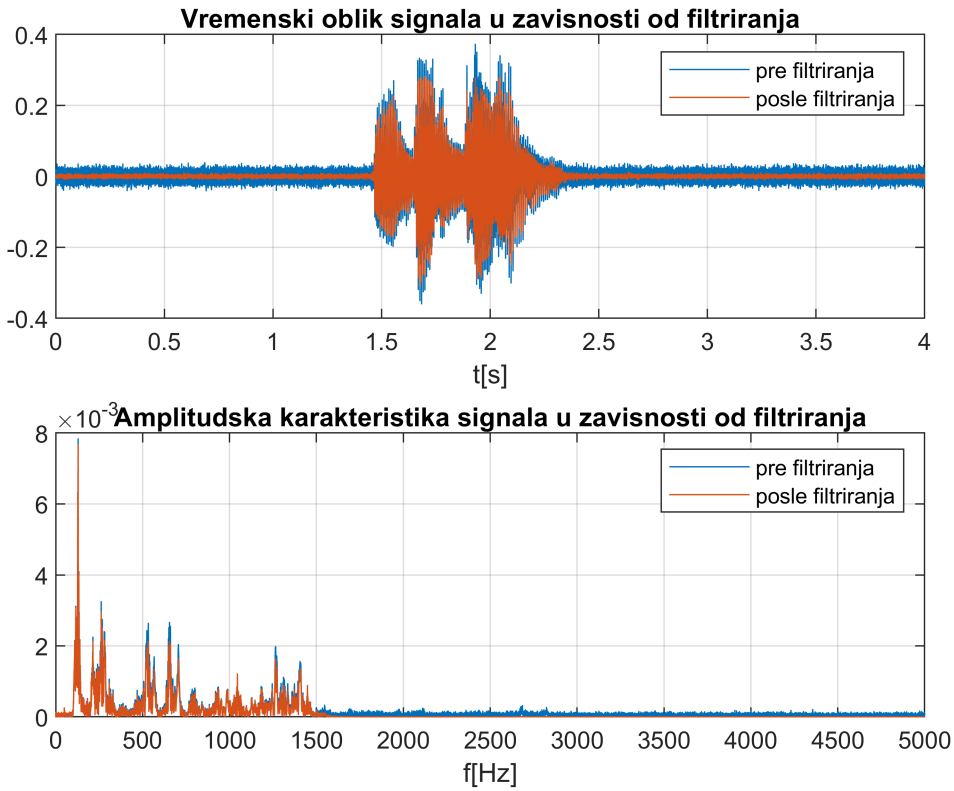
```
%-------------------------------------------------------
Wp = 1500 * 2*pi; %projektovanje analognog filtra
Ws = 2000 * 2*pi;
Rp = 2;
Rs = 40;
[n, Wn] = ellipord(Wp, Ws, Rp, Rs, 's');
[b, a] = ellip(n, Rp, Rs, Wp, 's');
[h, w] = freqs(b, a, N/2+1);

[bz, az] = impinvar(b, a, Fs); %diskretizovanje
[hz, fz] = freqz(bz, az, N/2+1, Fs);

y = filter(bz, az, y1); %filtriranje signala
Y = fft(y, N) / length(y);
Y1 = abs(Y(1 : N/2+1));
Y1(2 : N/2+1) = 2*Y1(2 : N/2+1);

%audiowrite('C:\Users\38166\Desktop\isfiltriran2.wav', y, Fs);
%Red filtra nam je 5.
%-------------------------------------------------------
figure(2); %amplitudska karakteristika filtra pre i posle diskretizacije
plot(w/(2*pi), 20*log10(abs(h)), 'LineWidth', 1.5); hold on;
plot(fz, 20*log10(abs(hz)), 'LineWidth', 1.5);
xlabel('f[Hz]'); grid on; legend('analogni', 'digitalni');
title('Amplitudska karakteristika filtra');
```




```
%-------------------------------------------------------
```

```
figure(3); %pre i posle dodavanja suma

subplot(2, 1, 1);
plot(t, y1); hold on;
plot(t, s);
xlabel('t[s]'); grid on; legend('sa sumom', 'bez suma');
title('Vremenski oblik signala u zavisnosti od dodavanja suma');

subplot(2, 1, 2);
plot(f1, X1); hold on;
xlim([0 5000]);
plot(fs, Xs1);
xlabel('f[Hz]'); grid on; legend('sa sumom', 'bez suma');
title('Amplitudska karakteristika signala u zavisnosti od dodavanja suma');
```




```
%-------------------------------------------------------
figure(4); %pre i posle filtriranja

subplot(2, 1, 1);
plot(t, y1); hold on;
plot(t, y);
xlabel('t[s]'); grid on; legend('pre filtriranja', 'posle filtriranja');
title('Vremenski oblik signala u zavisnosti od filtriranja');

subplot(2, 1, 2);
plot(f1, X1); hold on;
xlim([0 5000]);
plot(f1, Y1);
```

```
xlabel('f[Hz]'); grid on; legend('pre filtriranja', 'posle filtriranja');
title('Amplitudska karakteristika signala u zavisnosti od filtriranja');
```